

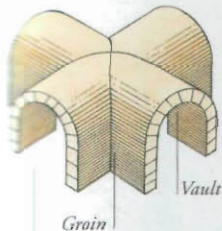
ARCHITECTURE



FROM A TOWERING SKYSCRAPER to a functional factory, architecture is the art of planning a building. The word also refers to the different building styles seen throughout history. Looking at changes in architecture tells us about earlier societies: the materials that were available to their builders, the skills mastered by their engineers, and the social ideals that they wished to express in their public buildings.

Architectural features

The main structural and functional features of a building are the roof, arches and walls, doors, and windows. The architect combines the practical knowledge of how to construct these with a sense of how to combine shape, space, and light to suit the function of the building itself.



Groin
Groin vault,
where two barrel
vaults intersect



Barrel vault
Round arch

Arch and vault

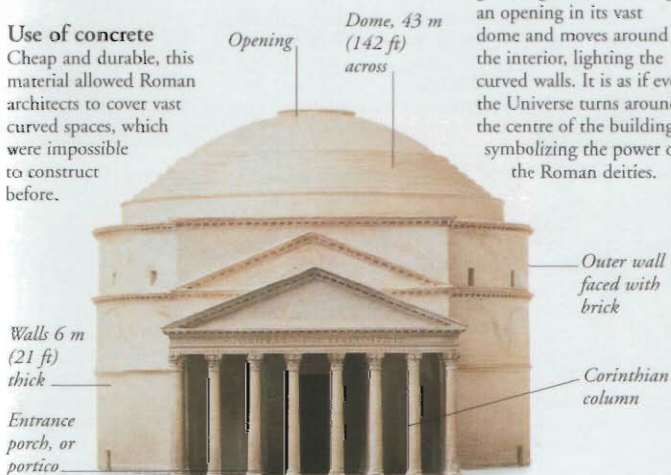
An arch is a curved or pointed structure that bridges a gap; it must carry the weight of the wall, floor, or roof above, and its structure allows it to support greater weight than a flat slab can. A vault is simply an arched ceiling.

Classical Europe

Classical architecture is that of the ancient Greeks and Romans. Both built by laying stones on top of each other, or by resting beams on columns. The Romans also developed the arch, vault, dome, and the use of concrete to develop curved spaces.

Use of concrete

Cheap and durable, this material allowed Roman architects to cover vast curved spaces, which were impossible to construct before.



The Pantheon, Rome, Italy, completed c. AD 128

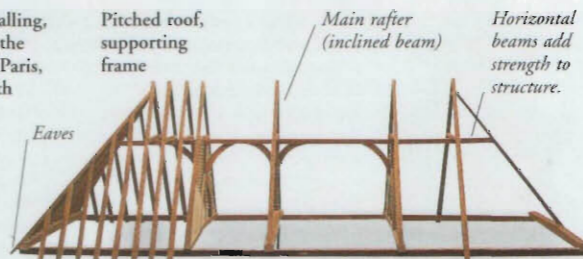


Dome metalling,
Church of the
Sorbonne, Paris,
France, 17th century

Cross
and orb
Lantern
(turret with
windows)
provides
light

Round-
arched
window

Dome on
a circular
base



Pitched roof,
supporting
frame

Main rafter
(inclined beam)

Horizontal
beams add
strength to
structure.

Dome

Domes – curved, solid roofs – were first built on palaces and religious buildings as striking symbols of the building's status. They are often difficult to build, and have been constructed in various shapes: the Dome of the Rock in Jerusalem is hemispherical; the "onion"-shaped dome is a popular feature of many Russian and Bavarian buildings.

Roof

All roofs are designed for the practical purpose of providing protection from the weather. The design and covering used will reflect the local climate: for instance, in a wet country a sloping (pitched) roof will let rain run off. Roofs can also be ingenious and beautiful, such as when crowning an ornate castle.

Brunelleschi

Italian architect Filippo Brunelleschi (1377–1446) returned to the use of Classical features, rejecting the Gothic style. Architects all across Europe followed his example.

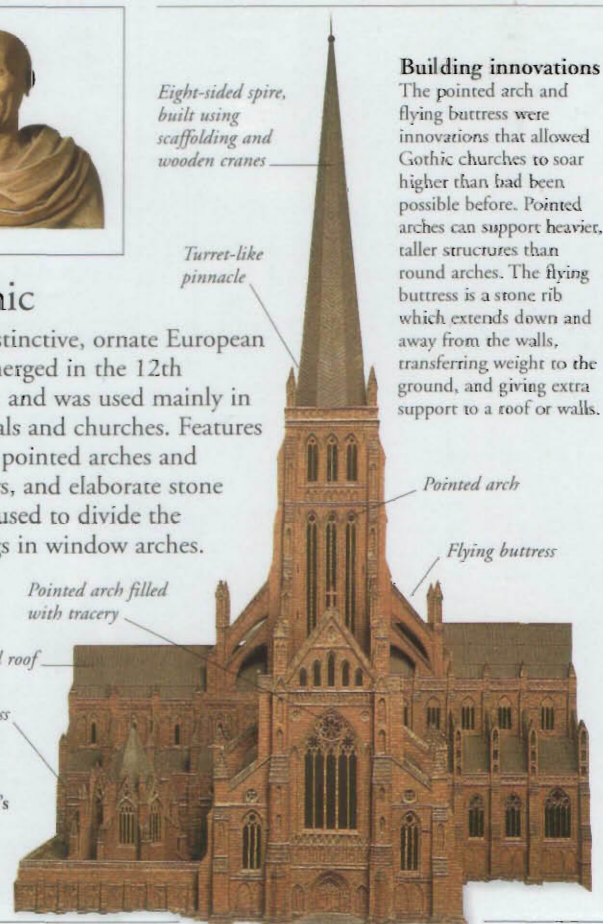


Symbolism

The Pantheon is a temple built to all the Roman gods. Light comes through an opening in its vast dome and moves around the interior, lighting the curved walls. It is as if even the Universe turns around the centre of the building, symbolizing the power of the Roman deities.

Gothic

This distinctive, ornate European style emerged in the 12th century, and was used mainly in cathedrals and churches. Features include pointed arches and windows, and elaborate stone tracery used to divide the openings in window arches.



Eight-sided spire,
built using
scaffolding and
wooden cranes

Turret-like
pinnacle

Building innovations

The pointed arch and flying buttress were innovations that allowed Gothic churches to soar higher than had been possible before. Pointed arches can support heavier, taller structures than round arches. The flying buttress is a stone rib which extends down and away from the walls, transferring weight to the ground, and giving extra support to a roof or walls.

Pointed arch

Flying buttress

Pointed arch filled
with tracery

Pitched roof

Buttress

Old St. Paul's
Cathedral,
London,
England,
1087–1666

Ornament

Early in the 20th century, many Western architects rejected all forms of building ornament. This is rare: most buildings from other periods and cultures use it extensively, and even a simple building will usually have some decoration to reflect the taste of its owner. The ancient Greeks, for instance, carved the tops, or capitals, of columns to dignify their most prestigious buildings. The distinct decorations were based on styles called orders.



Doric order



Ionic order



Corinthian order